Some Basic Concepts and Results of Probability Theory

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References

- R. Wolff: Sections 1-9, 1-14
- S.M. Ross, “Stochastic Processes”: Section 1.6

Indicator Random Variables

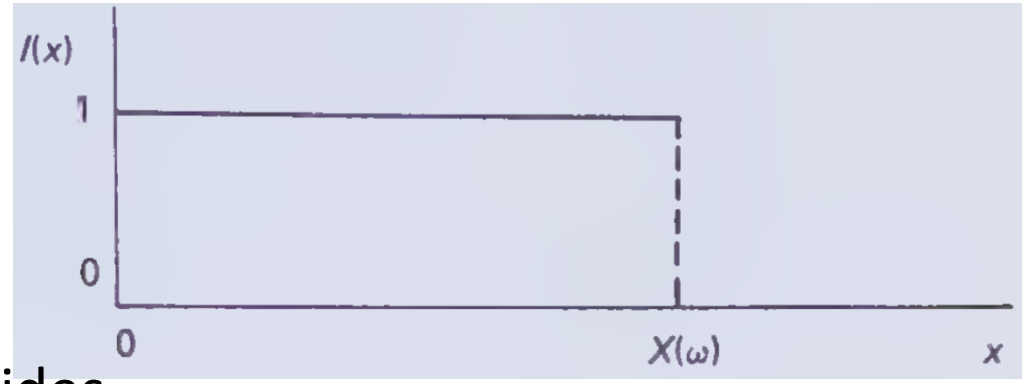
- Let A be any event
  - e.g., in a die roll experiment, the event that an odd number shows up
- $1_A = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{else.} \end{cases}$
  - indicator random variable corresponding to event A
- Distribution of 1_A:
  - Bernoulli($P(A)$)
- E(1_A):
  - $P(A)$

Random Variables, CDF, and Expectation

- Let X be a r.v. with CDF $F(x)$
- Often convenient to work with:
  - ❑ $F^c(x) := P(X > x) = 1 - F(x)$
  - ❑ known as *complementary CDF* (CCDF), *tail distribution*, or *survival function*
  - ❑ sometimes we use notation $\bar{F}(x)$ instead of $F^c(x)$
- Describes probability that component or system will survive beyond time x
- If X is a non-negative r.v., i.e., $P(X < 0) = 0$, but is otherwise arbitrary, then:

  1) $E(X) = \int_0^\infty F^c(x)dx$

- **Proof**:
  - ❑ $\int_0^\infty 1_{\{X>x\}}dx = X$ (see fig.)
  - ❑ Take expectations on both sides



Ref: R.W. Wolff, "Stochastic Modeling and the Theory of Queues"

Random Variables, CDF, and Expectation (contd.)

- Recall: if X is a non-negative r.v., then:

  1) $E(X) = \int_0^\infty F^c(x)dx$

- If X is an arbitrary r.v., then let:
  - ❑ $X = X^+ - X^-$,
  - ❑ where $X^+ = \max(X, 0)$ and $X^- = -\min(X, 0)$ are called positive and negative parts of X, respectively
- Hence:

  2) $E(X) = E(X^+) - E(X^-)$

- $E(X)$ is well-defined, but possibly infinite, if $E(X^+)$ or $E(X^-)$ is finite
- **Claim**:

  3) $E(X) = \int_0^\infty F^c(x)dx - \int_{-\infty}^0 F(x)dx$

- **Proof**:

  4) Note that $\int_{-\infty}^0 P(X \leq x)dx = \int_{-\infty}^0 P(X < x)dx$
  - ○ since areas under integrands of LHS and RHS are same
  - ❑ 3) follows from 1), 2), and 4)

Random Variables, CDF, and Expectation (contd.)

- Recall:
  1) $E(X) = \sum_x x p_x$ if X is discrete with PMF p_x
  2) $E(X) = \int_{-\infty}^{\infty} x f(x) dx$ if X is continuous with PMF $f(x)$
- $E(X)$ for mixed random variables can be calculated using 2) if impulse functions used
- Often convenient to combine 1) and 2) and write:
  3) $E(X) = \int_{-\infty}^{\infty} x dF(x)$
    - where $F(x)$ is the CDF of X
- Throughout this course, we regard 3) as concise notation to combine discrete, continuous, and mixed cases
- Expectation of arbitrary random variable X can be calculated using *Lebesgue-Stieltjes integral* for which notation in 3) used:
  - details omitted

Using Indicator Random Variables to Calculate Moments

- Recall: if X is a non-negative r.v., then:

  1) $E(X) = \int_0^\infty F^c(x)dx$

- E.g.:
  - ❑ Suppose X is exponential with rate μ, i.e.:
    - ○ $f_X(x) = \begin{cases} \mu e^{-\mu x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$
  - ❑ $P(X > x)$:
    - ○ $e^{-\mu x}$
  - ❑ By 1), $E(X)$:
    - ○ $\frac{1}{\mu}$
- 1) is valid for discrete as well as continuous r.v.s X
- **Exercise**: Show that if X is a non-negative integer-valued r.v., then:
  - ❑ $E(X) = \sum_{j=0}^\infty P(X > j)$

Using Indicator Random Variables to Calculate Moments (contd.)

- Recall: if X is a non-negative r.v., then:

  1) $E(X) = \int_0^\infty F^c(x)dx$

- Let X and Y be non-negative r.v.s with arbitrary joint distribution
- Then:

  2) $E(XY) = \int_0^\infty \int_0^\infty P(X > x, Y > y)\, dxdy$

- **Proof**:

  - ❑ $\int_0^\infty \int_0^\infty 1_{\{X>x,Y>y\}} dxdy = XY$
  - ❑ Take expectations on both sides

Using Indicator Random Variables to Calculate Moments (contd.)

- Recall: if X is a non-negative r.v., then:

  1) $E(X) = \int_0^\infty F^c(x)dx$

- Let X be a non-negative r.v. with CDF $F(.)$ and r be a positive integer
- Applying 1) to the r.v. X^r, we get $E(X^r)$:
  - ❑ $\int_0^\infty P(X^r > x)dx$
  - ❑ $\int_0^\infty P\left(X > x^{1/r}\right)dx$
  - ❑ $\int_0^\infty ru^{r-1}P(X > u)du$, by letting $x = u^r$
  - ❑ $r\int_0^\infty u^{r-1}F^c(u)du$

Hazard Rate Function

- Let X be a non-negative continuous r.v. with CDF $F(.)$ and PDF $f(.)$
- *Hazard rate function* or *failure rate function* of X:
  - $\lambda(t) = \frac{f(t)}{F^c(t)}$
- Interpretation:
  - Let X be lifetime of some item
  - Suppose it has survived for t hours
  - Probability that it will not survive for additional time dt is $P(X \in (t, t+dt) | X > t)$:
    - $\frac{P(X \in (t,t+dt), X>t)}{P(X>t)}$
    - $\frac{P\big(X \in (t,t+dt)\big)}{P(X>t)}$
    - $\approx \frac{f(t)dt}{F^c(t)}$
    - $\lambda(t)dt$
  - Thus, $\lambda(t)$ represents probability density that a t-year old item will fail

Hazard Rate Function (contd.)

- Recall:
  - Let X be a non-negative continuous r.v. with CDF $F(.)$ and PDF $f(.)$
  - Hazard rate function of X:
    - $\lambda(t) = \frac{f(t)}{F^c(t)}$
- Let the lifetime X be exponential with rate μ
- $\lambda(t)$:
  - $\frac{\mu e^{-\mu t}}{e^{-\mu t}}$
  - μ
- Interpretation:
  - By memoryless property, distribution of remaining life for a t-year-old item is same as for a new item
  - Hence, $\lambda(t)$ is constant